FACULTY OF ENGINEERING AND TECHNOLOGY DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

ENCS4310, Digital Signal Processing

| Assignment 1|

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Section: 1

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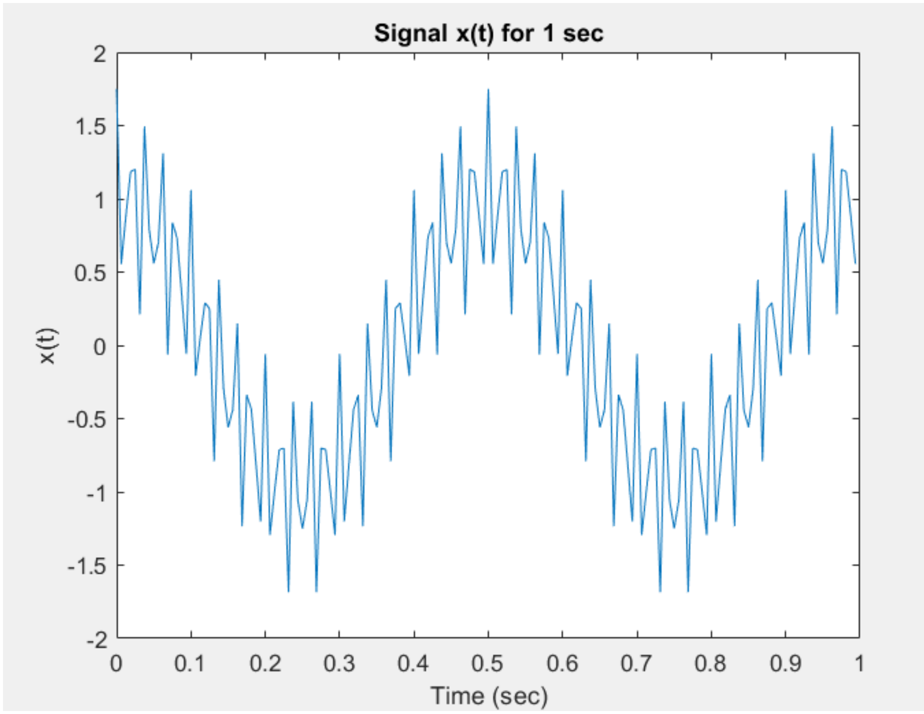
- **The code :**

```matlab
%aya dahbour 1201738
%A
%Define x(t)
t = 0:1/160:1-1/160; % 1 sec with Fs=160
x = cos(2*pi*2*t) + 0.5*cos(2*pi*50*t) + 0.25*cos(2*pi*80*t);

% Plot x[n]
n = 0:159; % Sample index
figure;
stem(n, x);
xlabel('Time (sec)');
ylabel('x(t)');
title('Signal x(t) for 1 sec');

%B
%Define the moving average filter function
MAF = @(M) ones(1,M+1)/(M+1);

%Compute and plot the filter frequency response for M = 1, 4, 10
M_values = [1 4 10];
w = linspace(-pi, pi, 1000);
figure;
for i = 1:length(M_values)
    M = M_values(i);
    H = freqz(MAF(M), 1, w);
    plot(w/pi, abs(H));
    hold on;
end
xlabel('Frequency (normalized)');
ylabel('|H(e^{jw})|');
title('Filter frequency response for different window sizes');
legend('M=1', 'M=4', 'M=10');

%C
%Compute and plot the output sequence y[n] for M = 1, 4, 10
figure;
for i = 1:length(M_values)
    M = M_values(i);
    y = conv(x, MAF(M), 'same'); % Apply the filter
    figure;
    plot(n, y);
    xlabel('n');
    ylabel('y[n]');
    title(sprintf('Output sequence y[n] for M=%d', M));
end

%D
%Define the window sizes and sampling frequency
M_values = [1 4 10];
fs = 160;
```

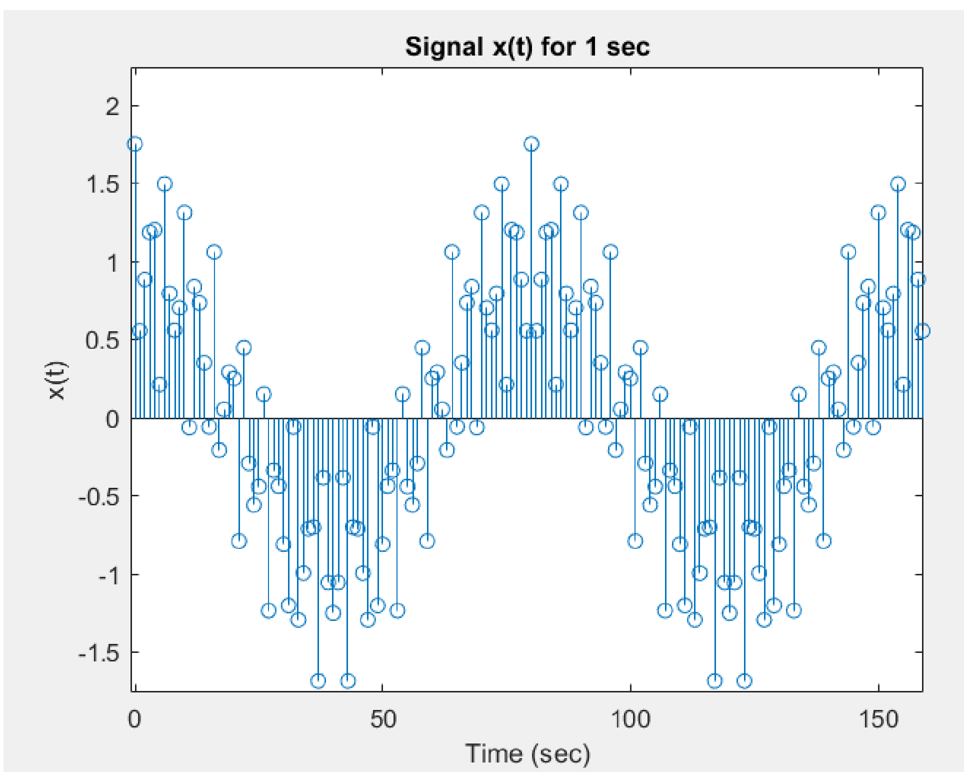
```
% plot the frequency spectra of x[n] and y[n]
X = fft(x);
f = (0:length(x)-1)*fs/length(x);

figure;
subplot(2,1,1);
plot(f, abs(X));
xlabel('Sample index');
ylabel('|X(e^{j\omega})|');
title('Magnitude spectrum of x[n]');

%Compute and plot the frequency spectra of y[n] for M = 1, 4, 10
figure;
for i = 1:length(M_values)
    M = M_values(i);
    y = conv(x, MAF(M), 'same'); % Apply the filter
    Y = fft(y);

    subplot(length(M_values), 1, i);
    plot(f, abs(Y));
    xlabel('Sample index');
    ylabel(sprintf('|Y(e^{j\\omega})| for M=%d', M));
    title(sprintf('Magnitude spectrum of y[n] for M=%d', M));
end

%E
%Define the window sizes and sampling frequency
M_values = 1:50;
fs = 160;

%Compute the MSE for different values of M
MSE = zeros(length(M_values), 1);
for i = 1:length(M_values)
M = M_values(i);
y = conv(x, MAF(M), 'same'); % Apply the filter
MSE(i) = mean((y - cos(2*pi*2*t)).^2); % Compute the MSE
end

%Plot the MSE versus M
figure;
plot(M_values, MSE);
xlabel('Window size M');
ylabel('MSE');
title('Mean squared error versus window size M');

%Find the minimum MSE and the corresponding M
[min_MSE, min_index] = min(MSE);
opt_M = M_values(min_index);
fprintf('The optimum window size to obtain the first sinusoidal signal
is M=%d with MSE=%f.\n', opt_M, min_MSE);
```

- **The results:**

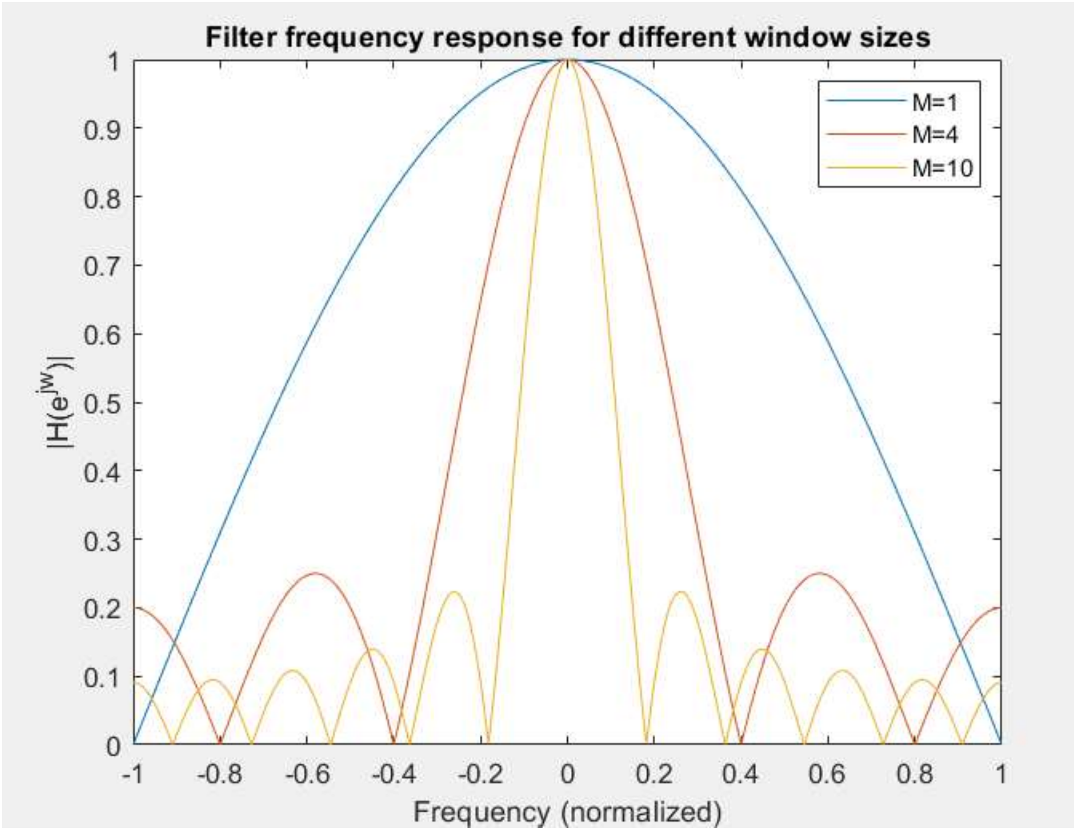
a) Using plot :



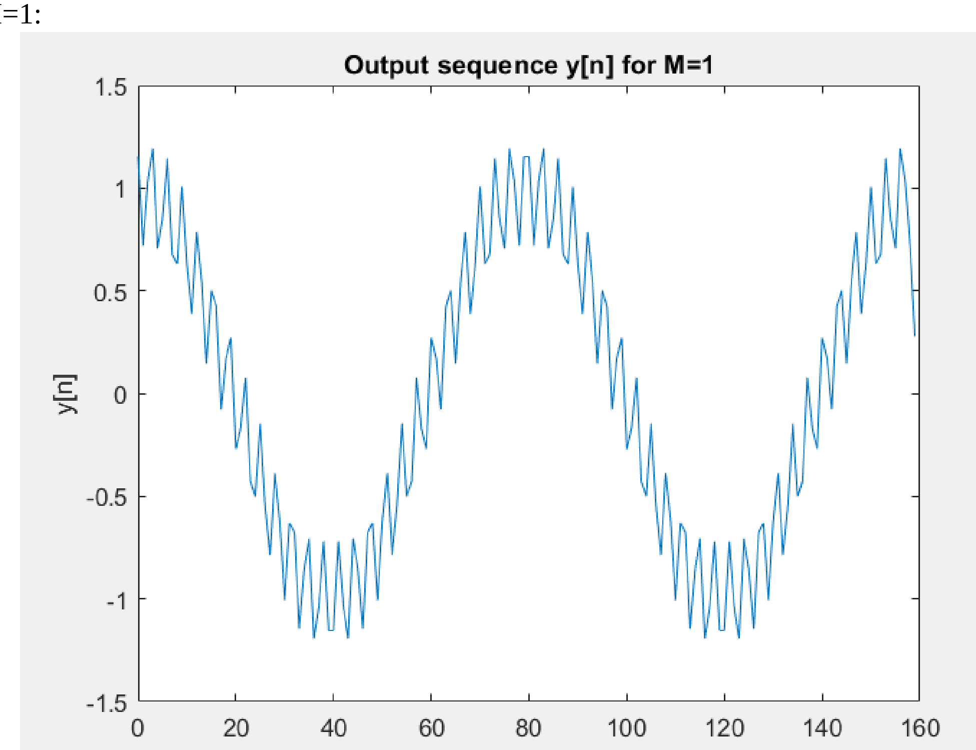
Using stem :



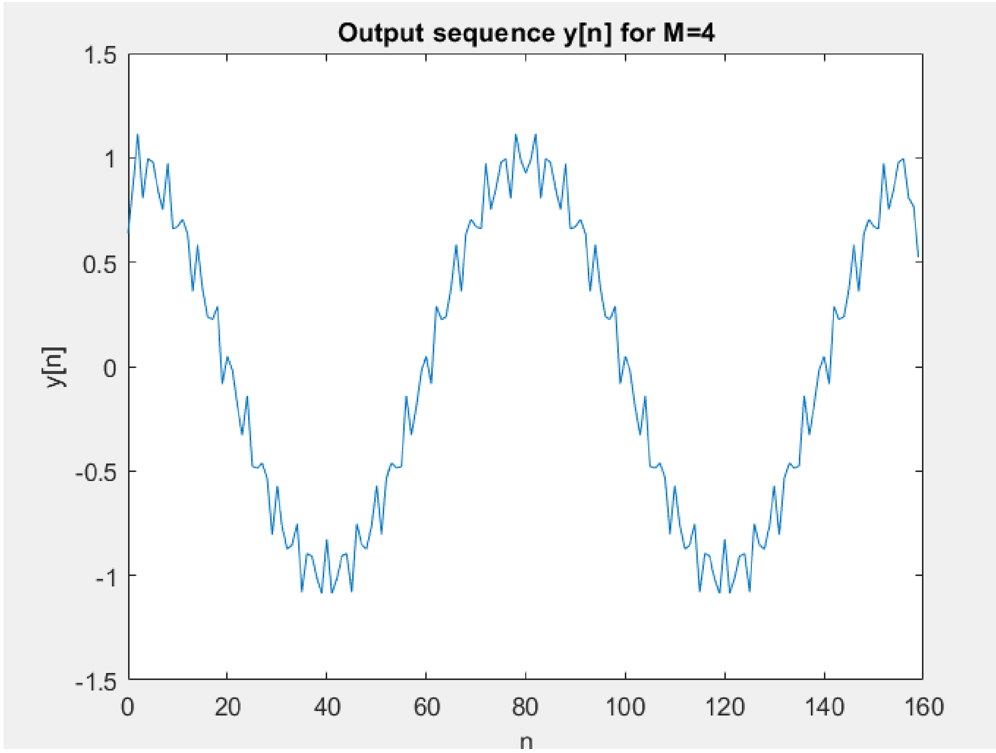
b)


Filter frequency response for different window sizes
M=1
M=4
M=10
$|H(e^{jw})|$
Frequency (normalized)
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
-1
-0.8
-0.6
-0.4
-0.2
0
0.2
0.4
0.6
0.8
1

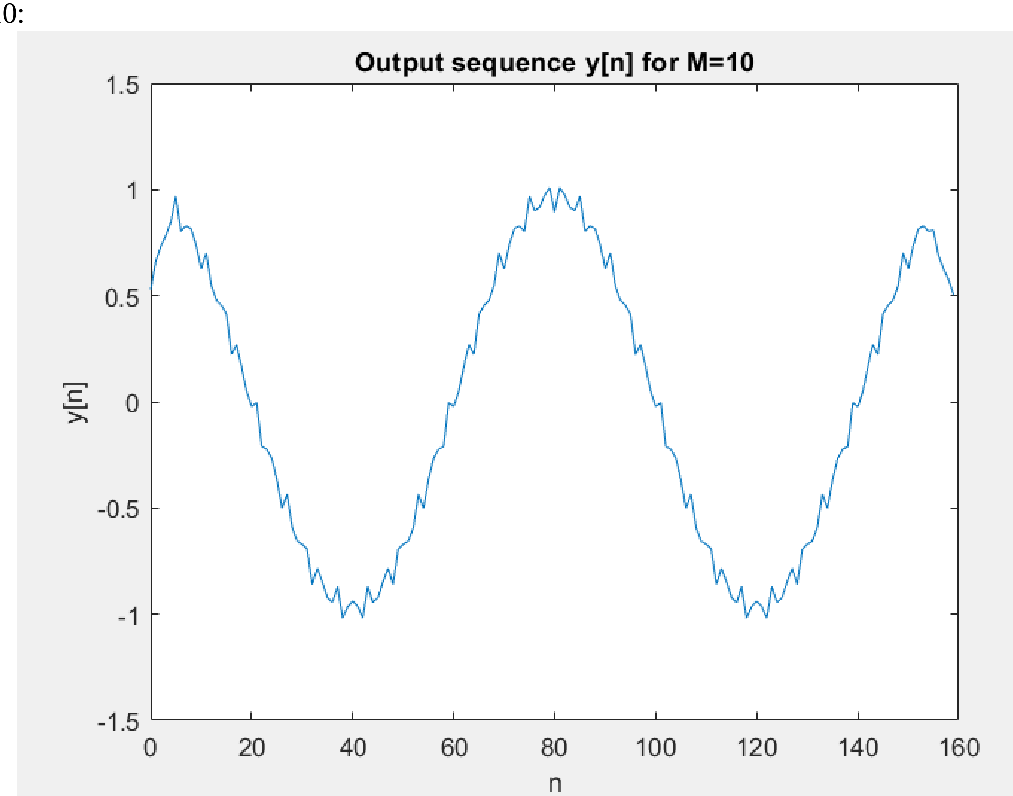
c) M=1:


Output sequence y[n] for M=1
y[n]
n
1.5
1
0.5
0
-0.5
-1
-1.5
0
20
40
60
80
100
120
140
160

M=4:

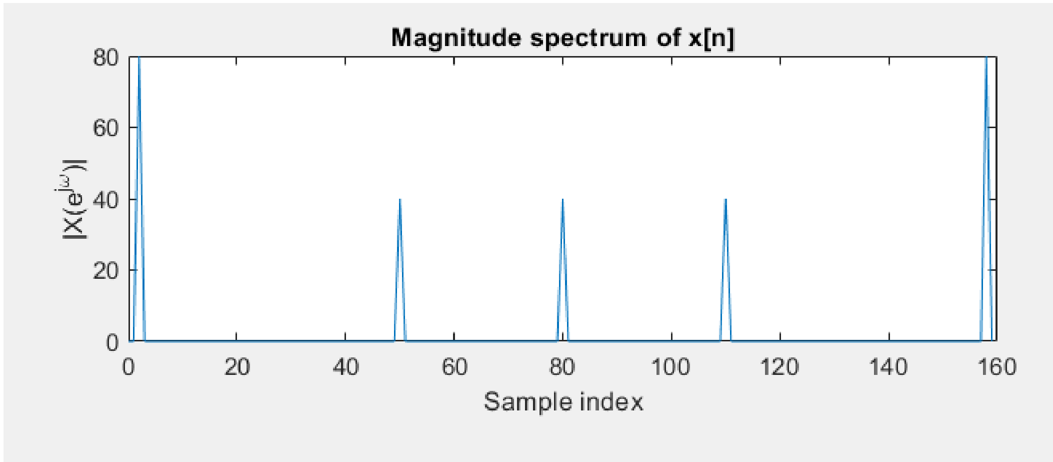

Output sequence y[n] for M=4
1.5
1
0.5
0
-0.5
-1
-1.5
y[n]
0
20
40
60
80
100
120
140
160
n

M=10:

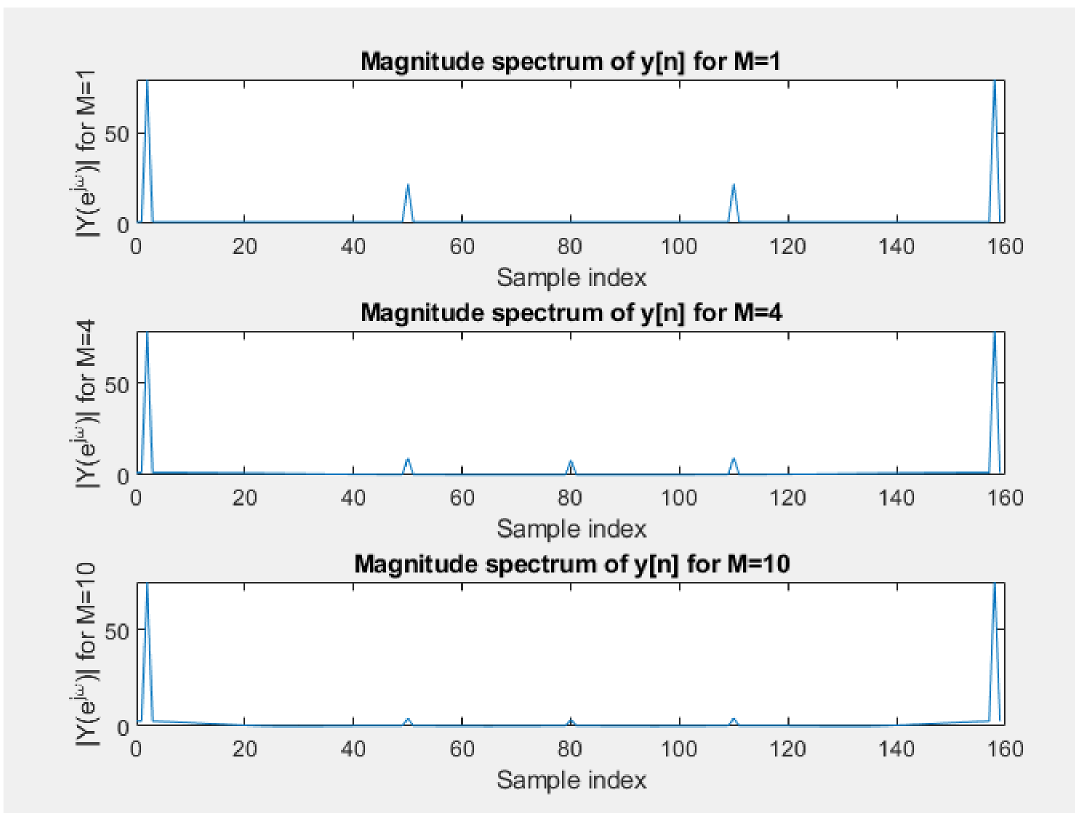

Output sequence y[n] for M=10
1.5
1
0.5
0
-0.5
-1
-1.5
y[n]
0
20
40
60
80
100
120
140
160
n

d)

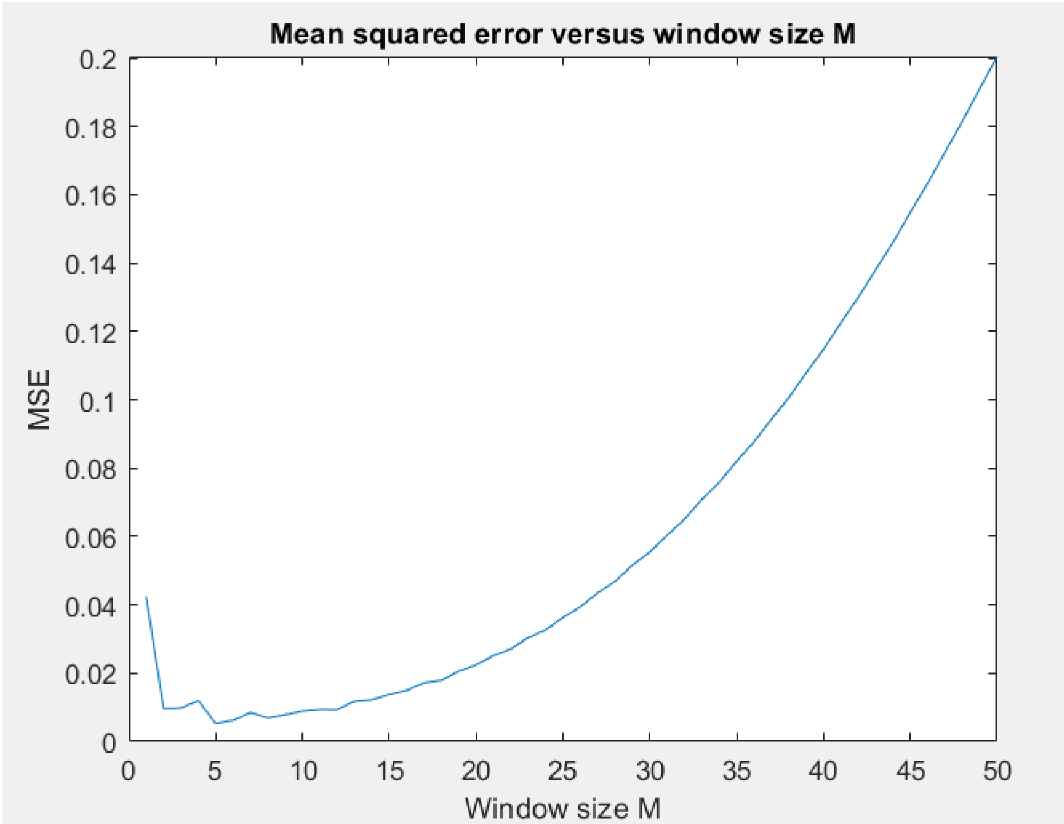
The input signal frequency spectrum:



The output frequency spectrum:



e)




```
>> dsp1
The optimum window size to obtain the first sinusoidal signal is M=5 with MSE=0.005289.
```